

الرياضيات المتقطعة

المحاضرة الثانية
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📌 Lecture: Counting Techniques in Discrete Mathematics

1. Introduction

Counting is one of the most fundamental concepts in combinatorics and discrete mathematics.

It allows us to determine how many possible outcomes or arrangements exist in a given situation — without having to list them all.

Counting is essential in computer science, probability, and logic, for example:

- Counting the number of possible passwords.
- Counting possible routes in a network.
- Counting combinations in data arrangements.

2. Basic Principles of Counting

2.1 The Addition Rule (Sum Rule)

If an event **A** can occur in m ways, and a disjoint event **B** can occur in n ways, then either **A** or **B** can occur in $m + n$ ways.

◆ *Disjoint means the events cannot happen at the same time.*

Example 1:

If a student can take **3 math courses** or **2 computer science courses**, and cannot take both, then there are $3 + 2 = 5$ choices of courses.

2.2 The Multiplication Rule (Product Rule)

If an operation consists of a sequence of steps,
and the first step can be done in m ways, and the second in n ways,
then the entire operation can be done in $m \times n$ ways.

Example 2:

If you have 3 shirts and 4 pants,
you can create $3 \times 4 = 12$ different outfits.

3. Factorials

A factorial, written as $n!$, represents the number of ways to arrange n distinct objects.

$$n! = n \times (n - 1) \times (n - 2) \times \cdots \times 2 \times 1$$

Example 3:

The number of ways to arrange 4 books on a shelf:

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

4. Permutations التباديل

4.1 Definition

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A permutation is an **ordered arrangement** of objects.

Order matters in permutations.

4.2 Formula

If we select r objects from n distinct ones:

$$P(n, r) = \frac{n!}{(n - r)!}$$

n: كبيرة
r: صغيرة

Example 4:

How many 3-letter "words" can be formed from the letters A, B, C, D?

$$P(4, 3) = \frac{4!}{(4 - 3)!} = \frac{24}{1} = 24$$

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4.3 Permutations with Repetition

If some elements repeat, divide by the factorial of repeated counts.

Example 5:

Number of distinct permutations of the word LEVEL:

- Total letters = 5
- Repeated letters: L (2), E (2)

$$\text{Permutations} = \frac{5!}{2! \times 2!} = \frac{120}{4} = 30$$

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5. Combinations

5.1 Definition

A combination is a selection of items without regard to order.

Order does not matter in combinations.

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5.2 Formula

$$C(n, r) = \frac{n!}{r!(n-r)!}$$

Example 6:

How many ways can we choose 3 students from a group of 10?

$$C(10, 3) = \frac{10!}{3!7!} = 120$$

5.3 Relation Between Permutations and Combinations

$$P(n, r) = C(n, r) \times r!$$

This means:

Note:-

If we know the number of combinations, multiplying by $r!$ gives the number of permutations.

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6. The Principle of Inclusion-Exclusion (PIE)

Used when counting elements in the union of overlapping sets.

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Example 7:

In a class of 50 students: n

- 30 study math r
- 25 study computer science
- 10 study both n

Number of students studying math or computer science:

$C(n, r)$

$$|A \cup B| = 30 + 25 - 10 = 45$$

7. Summary Table

Concept	Order Matters?	Formula	Example
Addition Rule	No	$m + n$	Course choices
Multiplication Rule	No Yes	$m \times n$	Shirts $\times$ Pants
Permutations	Yes	$P(n, r) = n! / (n - r)!$	Arranging letters
Combinations	No	$C(n, r) = n! / [r!(n - r)!]$	Selecting a team
Factorial	—	$n!$	Ordering all objects

8. Practice Problems

1. How many 4-digit numbers can be formed using digits 1–9 with no repetition?
2. In how many ways can a president, vice president, and secretary be chosen from 6 people?
3. How many different 3-flavor combinations can be made from 8 ice cream flavors?
4. From 20 students, how many committees of 5 can be formed?

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C(n, r)